

ECO101

Practice Sheet- Elasticity

Question-1

- a) When the price of oranges increases from \$1.00 per pound to \$1.50 per pound, quantity demanded falls from 500 pounds to 400 pounds. Calculate and interpret the price elasticity of demand.
- b) Calculate total revenue before and after the price change. How does that relate to the elasticity interpretation?
- c) Suppose consumer income increases from \$3,000 to \$3,300 per month. As a result, the quantity demanded for oranges increases from 400 pounds to 450 pounds. Calculate and interpret the income elasticity of demand (YED). Based on your result, is orange a normal good or an inferior good?

Question-2

Suppose,

$$Q_d = 1000 - 25P_i + 15P_c$$

Where:

Q_d = Quantity demanded of ice cream tubs

P_i = Price of ice cream tub (in dollars)

P_c = Price of cones (in dollars)

Assume,

$$P_i = 6 \text{ dollars}$$

$$P_c = 4 \text{ dollars}$$

- a) Calculate the initial quantity of ice cream tubs demanded.
- b) Suppose the price of ice cream tub falls to 5 dollars (while P_c is still \$4) Calculate and interpret the price elasticity of demand (PED).
- c) Now, suppose instead that the price of cones increases to 5 dollars (while P_i is again \$6). Calculate and interpret the cross-price elasticity of demand (XED).

Question-3

The table below shows the following information:

	Quantity demanded when income is \$10,000	Quantity demanded when income is \$20,000
Good-A	10	25
Good-B	4	5
Good-C	3	2

Using the information, calculate income elasticity of each good and interpret the values.

Question-4

Suppose the price of pizza increases from \$20 to \$30. As a result, the quantity supplied increases from 10 to 13 pizzas per hour.

- Calculate the price elasticity of supply (PES) using the midpoint method.
- Interpret your result. Is the supply elastic or inelastic? Explain why this might be the case based on the nature of the good.